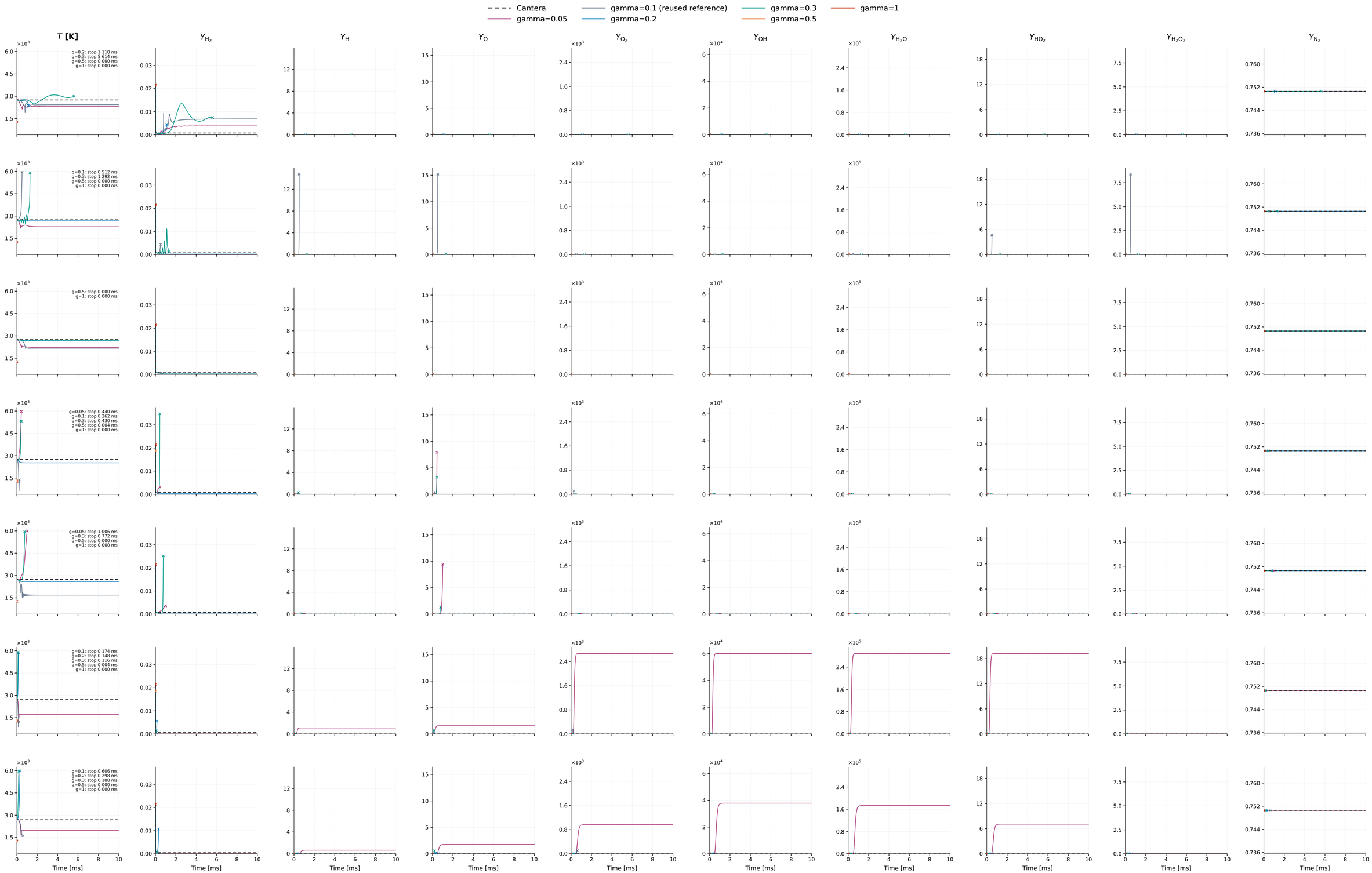
Box-Cox gamma sweep: Cantera vs NN | Val IC 00

T_0=1262.8 K P_0=4.913 atm ϕ_0=0.746 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 00

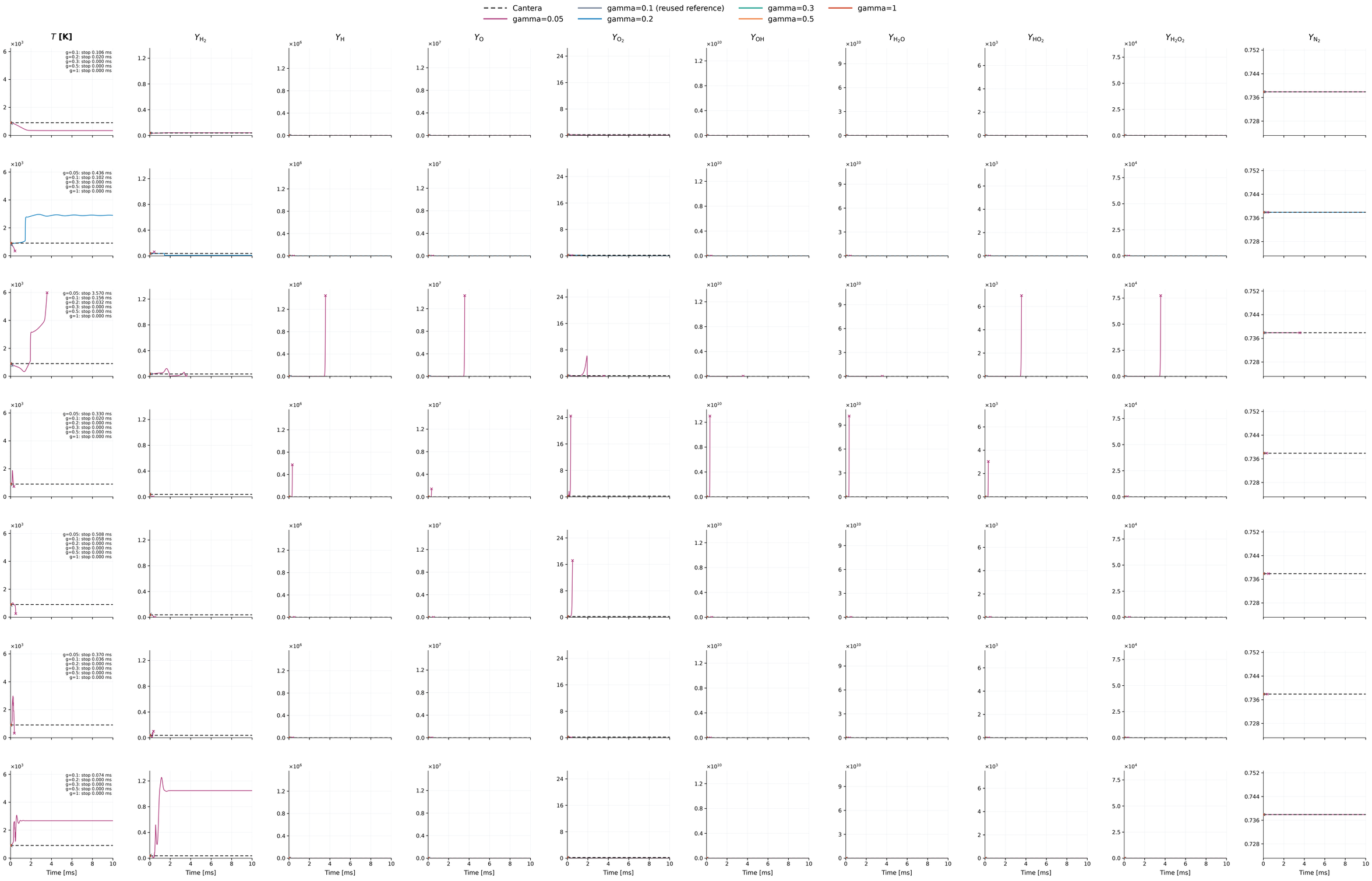
Raw | Cantera vs NN NRMSE | T0=1262.8K, P0=4.913atm, phi0=0.746

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.151	2.83	0.951	0.978	0.946	0.223	0.961	0.935	0.903	2.96e-16	Complete
1 ANN10	0.1	0.114	5.57	0.934	0.955	0.951	0.833	0.496	0.923	0.908	1.48e-16	Complete
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.118 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 5.614 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.165	0.67	0.952	0.732	1	4.07	0.494	0.195	0.159	2.96e-16	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.512 ms
2 P-only DeepONet10	0.2	0.0122	0.185	0.296	0.175	0.0474	0.143	0.0101	0.195	0.28	5.92e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 1.292 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.189	0.622	0.956	0.908	0.275	2.03	0.251	0.256	0.116	2.96e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	0.203	0.604	0.855	0.928	0.473	0.81	0.553	0.0975	0.268	2.96e-16	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0266	0.242	0.431	0.299	0.0684	0.239	0.0223	0.262	0.357	5.92e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0286	0.224	0.38	0.307	0.0917	0.205	0.0403	0.214	0.237	2.96e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.440 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.262 ms
4 T/P DeepONet10	0.2	0.0819	0.49	0.667	0.572	0.14	0.384	0.0567	0.35	0.251	5.92e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.430 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.006 ms
5 ANN10 + T/P DeepONet10	0.1	0.377	0.772	12.9	0.579	31.2	38.8	20.1	46.7	36	2.96e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0528	0.393	0.556	0.454	0.042	0.318	0.0191	0.341	0.261	5.92e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.772 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.367	0.779	8.96e+03	578	4.67e+04	4.19e+06	1.61e+06	7.46e+05	2.69e+03	2.96e-16	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.174 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.116 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.271	0.764	5.3e+03	644	1.65e+04	2.59e+06	9.45e+05	2.70e+05	2.27e+03	2.96e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.606 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.298 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.188 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 09

T_0=901.6 K P_0=2.084 atm ϕ_0=1.343 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 09

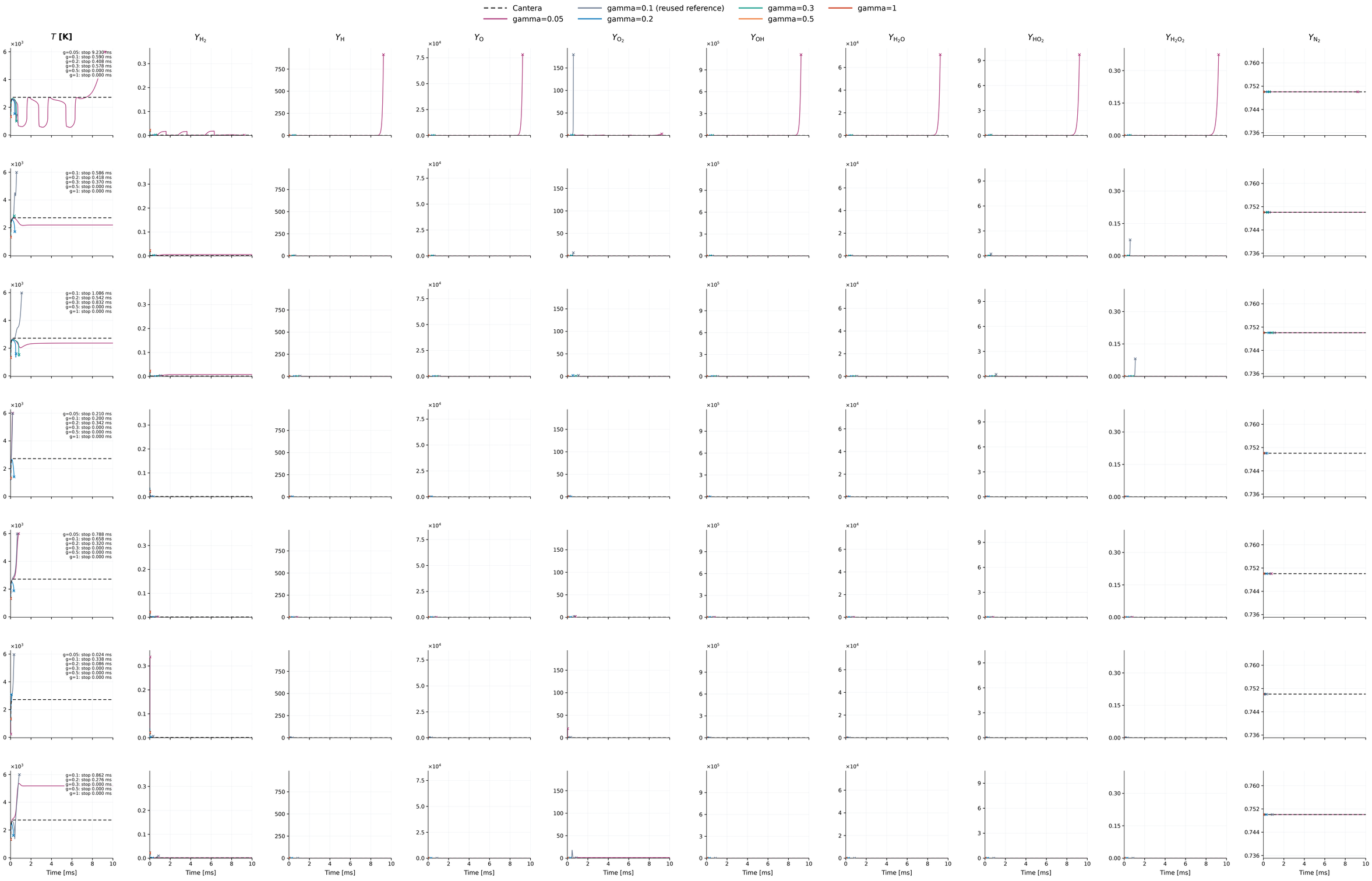
Raw | Cantera vs NN NRMSE | T0=901.6K, P0=2.084atm, phi0=1.343

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.591	0.102	1	0.999	0.631	1	1	1	1	7.52e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.106 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.102 ms
2 P-only DeepONet10	0.2	2.04	0.695	7.63e+08	1.16e+09	0.91	2.66e+10	9.53e+06	137	18	4.51e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 3.570 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.156 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.032 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.330 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.508 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.370 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	1.93	25.6	1.02e+12	2.19e+12	0.985	5.80e+12	3.22e+09	87	19.4	7.52e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 10

T_0=1317.2 K P_0=1.262 atm ϕ_0=0.769 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 10

Raw | Cantera vs NN NRMSE | T0=1317.2K, P0=1.262atm, phi0=0.769

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 9.230 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.590 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.408 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.578 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.186	2.28	11.5	0.161	0.794	9.74	3.51	1.36	1.97	8.88e-16	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.586 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.418 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.370 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.139	3.41	20.2	0.217	0.876	26.2	9.35	2.14	5.21	1.33e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.086 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.542 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.832 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.200 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.342 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.788 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.658 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.320 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.338 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.086 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.881	0.353	6.5	120	26.2	6.51	1.63	1.32	0.3	1.33e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.862 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.276 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 14

T_0=1043.0 K P_0=1.518 atm ϕ_0=0.520 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 14

Raw | Cantera vs NN NRMSE | T0=1043.0K, P0=1.518atm, phi0=0.520

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 5.104 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.348 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.044 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.628 ms
2 P-only DeepONet10	0.1	0.192	0.613	1.54	0.974	8.29	9.06	0.134	31.1	41.1	1.03e-15	Complete
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.228 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.550 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.754 ms
3 ANN10 + P-only DeepONet10	0.2	0.599	8.04	1	1	1.38	1	1	0.903	1.13	5.88e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	0.28	0.794	5.02e+11	2.88e+10	654	9.03e+13	4.57e+12	1.67e+07	4.08e+05	2.35e-15	Complete
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.200 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.176 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.754 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.572 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.276 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	1.49	5.32	21	1.65e+03	22	41.1	1.8	5.74	53.5	2.35e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.086 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 17

T_0=1057.2 K P_0=1.276 atm ϕ_0=0.692 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 17

Raw | Cantera vs NN NRMSE | T0=1057.2K, P0=1.276atm, phi0=0.692

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.523	3.1	5.55	14.9	11	6.26	0.782	12.1	27.6	8.86e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.314 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.536 ms
2 P-only DeepONet10	0.1	0.295	0.668	0.741	0.845	3.4	0.963	0.441	10.1	18.8	5.91e-16	Complete
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.416 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.534 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.916 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.290 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.276 ms
4 T/P DeepONet10	0.1	0.44	0.682	6.79e+06	3.06e+04	21.4	4.67e+06	6.08e+05	1.17e+04	114	5.91e-16	Complete
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.128 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.614 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.626 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.380 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.362 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.086 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.168 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 19

T_0=1416.7 K P_0=2.395 atm ϕ_0=0.972 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 19

Raw | Cantera vs NN NRMSE | T0=1416.7K, P0=2.395atm, phi0=0.972

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.583	5.41	0.926	0.965	7.7	0.895	0.933	0.851	9.28	1.94e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.194 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.914 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.510 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.618 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.682 ms
2 P-only DeepONet10	0.2	0.0515	0.242	0.44	0.505	0.285	0.295	0.0804	0.517	0.478	2.98e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.034 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.134 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.390 ms
3 ANN10 + P-only DeepONet10	0.2	0.0522	0.251	0.454	0.52	0.292	0.308	0.0742	0.513	0.48	1.49e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0482	0.339	0.511	0.414	0.123	0.273	0.0739	0.173	0.307	0	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	0.986	0.224	9.05	233	39.6	310	0.522	10.7	22.5	1.94e-15	Complete
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.584 ms
4 T/P DeepONet10	0.2	0.0271	0.137	0.291	0.306	0.183	0.191	0.0768	0.241	0.188	2.98e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.322 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.042 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.462 ms
5 ANN10 + T/P DeepONet10	0.2	0.0342	0.195	0.378	0.379	0.187	0.235	0.0834	0.259	0.185	1.49e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.830 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.337	0.961	251	604	4.91e+03	1.36e+06	7.04e+04	1.40e+05	3.67e+03	1.94e-15	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.324 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.216 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.116 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.162	709	9.8e+03	3.66e+03	0.949	1.12e+03	6.99e+03	8.87	264	2.08e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.522 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.776 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.260 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 23

T_0=1226.9 K P_0=1.917 atm ϕ_0=1.176 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 23

Raw | Cantera vs NN NRMSE | T0=1226.9K, P0=1.917atm, phi0=1.176

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 8.188 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.654 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 2.650 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.978 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.070 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.180 ms
2 P-only DeepONet10	0.2	0.0292	0.0986	0.262	0.629	0.314	0.382	0.0984	0.388	0.601	2.99e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.618 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.490 ms
3 ANN10 + P-only DeepONet10	0.2	4.54e-03	0.144	0.0382	0.104	0.111	0.152	0.0421	0.316	0.618	4.49e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.442 ms
4 T/P DeepONet10	0.1	4.19e-03	0.0409	0.0763	0.0927	0.122	0.0863	0.0212	0.265	0.166	5.99e-16	Complete
4 T/P DeepONet10	0.2	8.35e-03	0.105	0.152	0.123	0.278	0.0316	0.0612	0.5	0.216	2.99e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.178 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.724	0.946	2.67e+12	6.42e+09	1.65	9.16e+14	1.02e+15	8.58e+08	7.32e+09	1.35e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	4.04e-03	0.0283	0.0718	0.0952	0.101	0.0857	0.0222	0.187	0.144	1.05e-15	Complete
5 ANN10 + T/P DeepONet10	0.2	4.43e-03	0.0832	0.152	0.136	0.201	0.0409	0.0336	0.385	0.143	4.49e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.282 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.784	0.0746	2.57	91	28	5.88	0.914	2.9	2.65	1.80e-15	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.520 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.294 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.165	18.5	2.98e+03	12.8	0.581	397	322	18.1	244	1.35e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.954	2.31	4.4	550	61.8	210	0.496	22.9	114	1.05e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.564 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.334 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 24

T_0=1522.6 K P_0=1.432 atm ϕ_0=0.599 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 24

Raw | Cantera vs NN NRMSE | T0=1522.6K, P0=1.432atm, phi0=0.599

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.544	9.03	7.4	6.02	9.21	37.2	1.27	15.6	85.6	2.06e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 7.310 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.960 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.558 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.23	3.07	7.98	0.41	0.862	19.7	4.03	1.23	1.44	2.06e-15	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.548 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.630 ms
2 P-only DeepONet10	0.3	0.237	0.757	0.845	0.667	2.58	0.593	0.702	0.922	0.834	2.95e-16	Complete
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.182 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.976 ms
3 ANN10 + P-only DeepONet10	0.2	0.149	0.518	0.769	2.47	1.22	3.63	0.458	0.956	0.974	4.42e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.796 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.292 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.270 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.338 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.644 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.564 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.382 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.366 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.250 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.230 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	1.03	0.936	6.32	218	23.3	9.33	1.85	1.43	0.464	1.91e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.090 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.370 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 29

T_0=1127.4 K P_0=2.860 atm ϕ_0=1.850 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 29

Raw | Cantera vs NN NRMSE | T0=1127.4K, P0=2.860atm, phi0=1.850

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.327	9.77	1.07	3.36	0.331	24.9	0.815	0.919	5.26	6.10e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.786 ms
1 ANN10	0.2	0.118	0.142	0.919	1.13	0.4	0.451	0.288	0.9	1.04	1.53e-16	Complete
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 3.712 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.362 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.738 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.146 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.0831	1.07	1.19e+04	230	1.17	318	1.43e+03	882	3.93e+04	7.63e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.700 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.952 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.162 ms
4 T/P DeepONet10	0.1	0.0125	0.0332	0.197	0.593	0.476	0.115	0.038	1.53	1.81	4.58e-16	Complete
4 T/P DeepONet10	0.2	0.0202	0.039	0.283	0.742	0.546	0.222	0.0362	2.03	2.17	1.53e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.354 ms
5 ANN10 + T/P DeepONet10	0.1	0.0171	0.0359	0.405	0.635	0.26	0.355	0.219	0.694	0.902	6.10e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0391	0.0431	0.388	0.871	0.488	0.429	0.0751	1.45	1.52	7.63e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.198 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.244 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.358 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.972 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0358	0.275	0.777	1.35	0.334	1.07	0.242	0.948	1.12	6.10e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.828 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.086 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 33

T_0=1254.9 K P_0=4.275 atm ϕ_0=0.886 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 33

Raw | Cantera vs NN NRMSE | T0=1254.9K, P0=4.275atm, phi0=0.886

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.166	1.68	0.953	0.981	0.908	0.661	0.861	0.921	0.917	4.45e-16	Complete
1 ANN10	0.1	0.155	3.17	0.952	0.964	0.92	0.92	0.51	0.917	0.913	2.97e-16	Complete
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.018 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 4.976 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.528 ms
2 P-only DeepONet10	0.1	0.248	0.852	0.881	0.893	5.78	0.43	0.402	4.25	2.02	2.97e-16	Complete
2 P-only DeepONet10	0.2	0.0367	0.258	0.426	0.321	0.0726	0.243	0.0919	0.221	0.181	2.97e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.142 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.239	0.739	0.955	0.955	0.42	0.586	0.538	0.424	0.413	4.45e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	0.267	0.781	0.905	0.966	1.49	0.85	0.642	0.763	0.742	4.45e-16	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0457	0.303	0.503	0.376	0.0529	0.299	0.064	0.265	0.288	1.48e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0492	0.386	0.56	0.386	0.122	0.29	0.0463	0.0998	0.132	4.45e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.490 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.652 ms
4 T/P DeepONet10	0.2	0.0304	0.239	0.311	0.26	0.133	0.173	0.0714	0.334	0.217	2.97e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.522 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.980 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.164 ms
5 ANN10 + T/P DeepONet10	0.2	0.0371	0.256	0.388	0.335	0.135	0.234	0.0714	0.376	0.244	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.440 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.386	0.906	1.44e+03	232	2.02e+04	2.04e+06	4.63e+05	3.61e+05	3.88e+03	4.45e-16	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.110 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.309	0.896	430	141	4.33e+03	5.11e+05	1.05e+05	6.46e+04	1.7e+03	4.45e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.418	0.892	0.865	22.4	2.95e+03	693	394	166	5.32	4.45e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.230 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.168 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 41

T_0=1339.8 K P_0=1.702 atm ϕ_0=1.947 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10
2 P-only DeepONet10
3 ANN10 + P-only DeepONet10
4 T/P DeepONet10
5 ANN10 + T/P DeepONet10
6 T/P/ϕ DeepONet10
7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 41

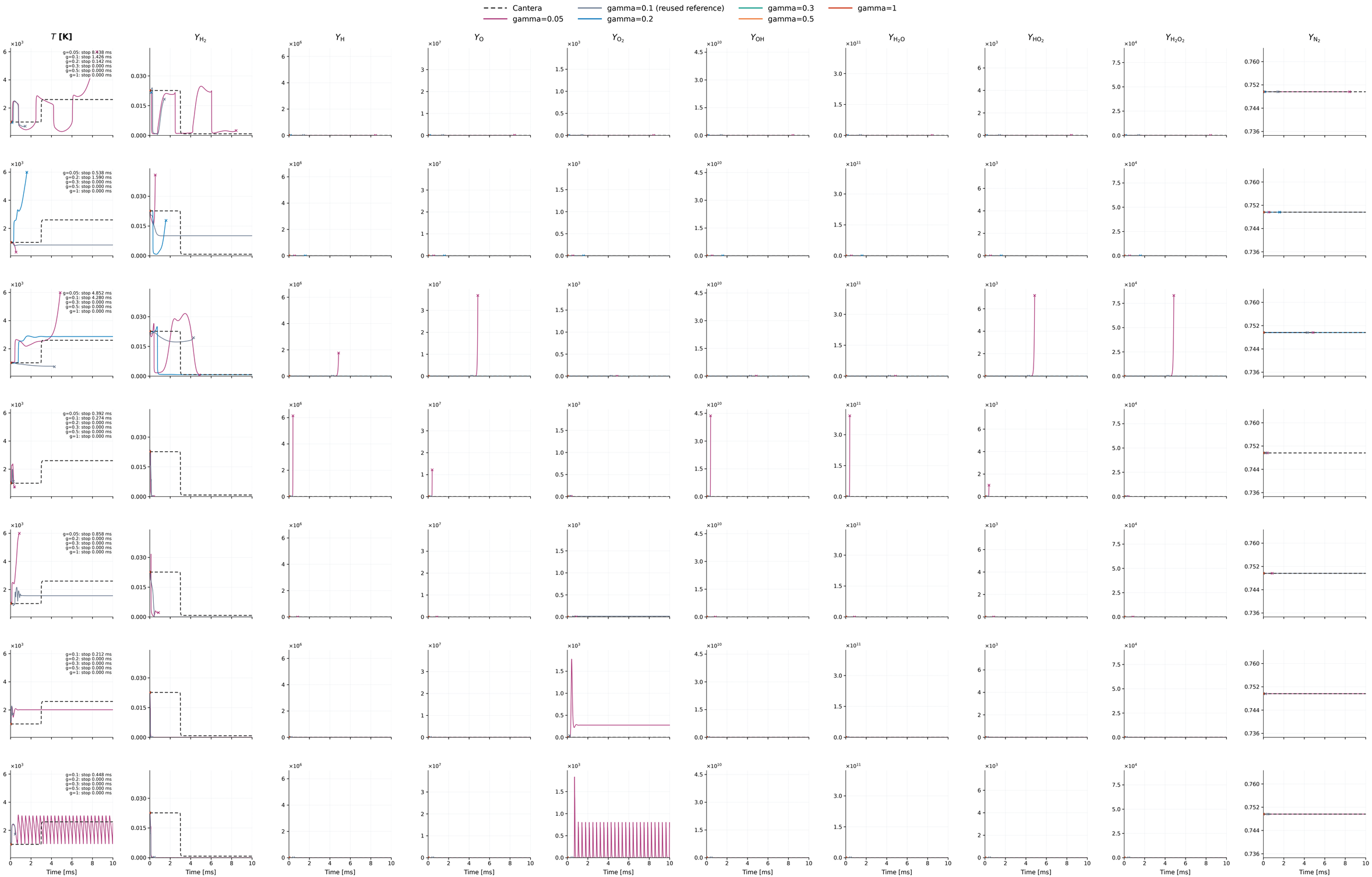
Raw | Cantera vs NN NRMSE | T0=1339.8K, P0=1.702atm, phi0=1.947

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.371	11.7	2.38	2.94	0.446	18	3.59	1.02	9.18	6.12e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.714 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 4.900 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.506 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.922 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.728 ms
2 P-only DeepONet10	0.2	0.0755	0.264	0.359	0.601	0.0475	0.521	0.0915	0.115	2	6.12e-16	Complete
2 P-only DeepONet10	0.3	0.0695	4.97e-03	0.463	0.595	0.0359	0.472	0.021	0.165	0.642	0	Complete
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.752 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 5.408 ms
3 ANN10 + P-only DeepONet10	0.2	0.0795	0.277	0.435	0.692	0.0508	0.616	0.11	0.0864	1.04	1.53e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 7.070 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.270 ms
4 T/P DeepONet10	0.1	0.028	0.0295	0.44	0.642	0.244	0.445	0.06	0.435	0.445	3.06e-16	Complete
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.0237	0.0263	0.306	0.133	0.157	0.17	0.209	0.271	0.249	1.53e-16	Complete
5 ANN10 + T/P DeepONet10	0.1	0.041	0.142	0.584	0.739	0.165	0.582	0.256	0.313	0.437	3.06e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.546 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.292 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
6 T/P/ϕ DeepONet10	0.2	0.0402	0.0208	0.304	0.518	0.349	0.36	0.0174	0.648	0.463	6.12e-16	Complete
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.440 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0478	0.0735	0.37	0.554	0.216	0.452	0.124	0.422	0.408	3.06e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0398	0.0148	0.337	0.532	0.253	0.38	0.012	0.501	0.425	1.53e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 46

T_0=1002.9 K P_0=1.808 atm ϕ_0=0.789 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 46

Raw | Cantera vs NN NRMSE | T0=1002.9K, P0=1.808atm, phi0=0.789

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 8.438 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.426 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.142 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.538 ms
2 P-only DeepONet10	0.1	0.665	0.806	1	1	2.18	1	1	0.984	0.96	2.96e-16	Complete
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.590 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 4.852 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 4.280 ms
3 ANN10 + P-only DeepONet10	0.2	0.387	0.829	1.65	2.85	0.644	1.94	0.56	1.08	1.03	1.48e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.392 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.274 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.858 ms
5 ANN10 + T/P DeepONet10	0.1	0.411	0.94	87.5	3.27	127	3.9e+03	944	275	8.2	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.326	0.987	5.28e+03	1.37e+04	2.46e+03	5.26e+06	9.60e+04	8.96e+04	229	2.96e-16	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.212 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.409	0.979	1.9e+03	1.39e+04	2.15e+03	2.91e+05	1.36e+04	3.64e+03	106	1.48e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.448 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 48

T_0=1440.4 K P_0=4.726 atm ϕ_0=1.757 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 48

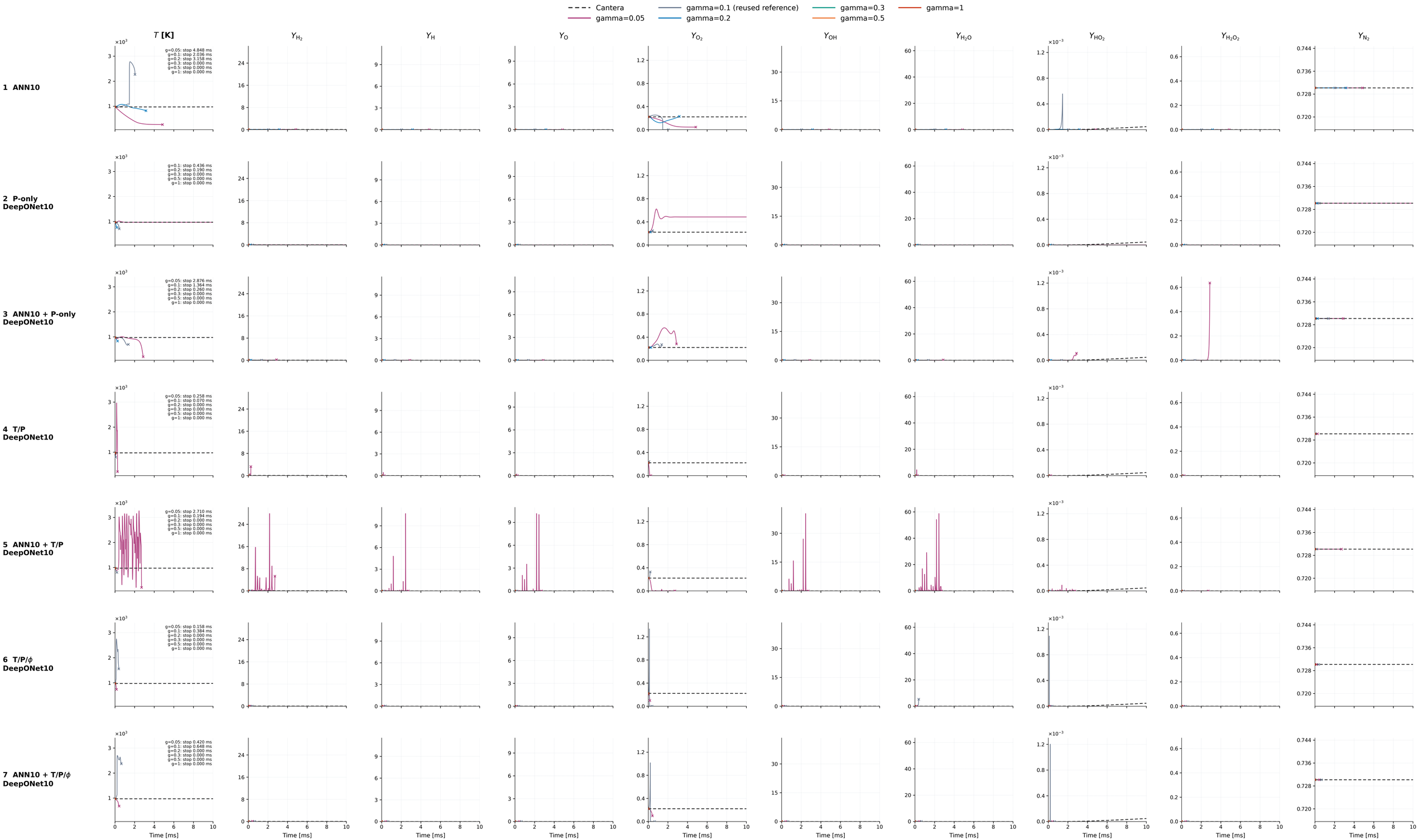
Raw | Cantera vs NN NRMSE | T0=1440.4K, P0=4.726atm, phi0=1.757

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.143	0.626	0.969	0.954	0.108	0.778	2.15	0.715	0.769	3.35e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.744 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 2.618 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 4.490 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.506 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.276 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.406 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.316 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 7.098 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.582 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.620 ms
3 ANN10 + P-only DeepONet10	0.3	0.075	0.0312	0.555	0.778	0.117	0.607	0.0624	0.482	0.392	3.04e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.034 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	0.0104	0.0215	0.12	0.163	0.116	0.164	0.076	1.08	0.276	3.35e-15	Complete
4 T/P DeepONet10	0.1	0.0533	9.34e-03	0.443	0.697	0.145	0.514	0.0412	0.891	0.6	4.57e-16	Complete
4 T/P DeepONet10	0.2	0.0701	7.59e-03	0.521	0.765	0.153	0.595	0.02	0.886	0.641	1.52e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.254 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.267	117	255	644	1.12	197	16.9	2.46	3.76	3.35e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0781	0.0792	0.695	0.883	0.131	0.735	0.198	0.657	0.709	9.13e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.818 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.496 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.278 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.596 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.126 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.074 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.192 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.600 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 52

T_0=972.8 K P_0=4.350 atm ϕ_0=1.718 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 52

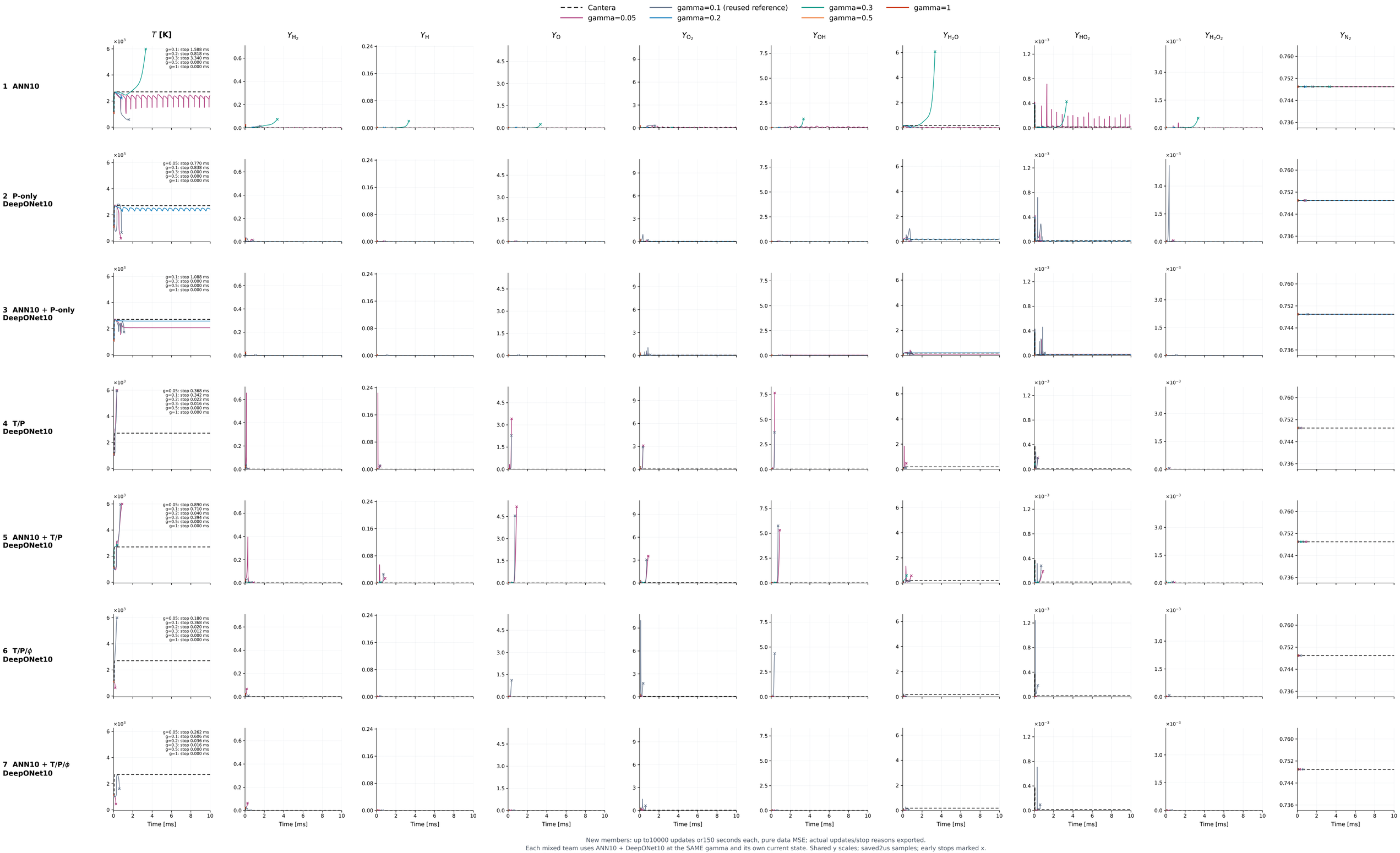
Raw | Cantera vs NN NRMSE | T0=972.8K, P0=4.350atm, phi0=1.718

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 4.848 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.036 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 3.158 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	8.75e-03	0.243	0.999	0.976	1.18	0.923	1	0.995	1	9.12e-16	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.190 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.876 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.364 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.260 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.258 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.710 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.194 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.158 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.384 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.420 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.648 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 63

T_0=1101.2 K P_0=3.120 atm ϕ_0=0.820 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 63

Raw | Cantera vs NN NRMSE | T0=1101.2K, P0=3.120atm, phi0=0.820

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.158	0.402	0.916	0.869	0.954	4.95	0.863	2.11	8	3.26e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.588 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.818 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 3.340 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.770 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.838 ms
2 P-only DeepONet10	0.2	0.0994	0.244	0.621	0.609	0.222	0.433	0.104	0.461	1.06	1.48e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.227	0.413	0.976	0.938	0.351	1.81	0.494	0.592	1.02	3.41e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.088 ms
3 ANN10 + P-only DeepONet10	0.2	0.0462	0.0749	0.374	0.473	0.228	0.329	0.0838	0.245	0.657	1.19e-15	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.368 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.342 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.022 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.890 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.710 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.040 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.394 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.368 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.262 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.606 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 71

T_0=1549.8 K P_0=1.296 atm ϕ_0=1.887 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 71

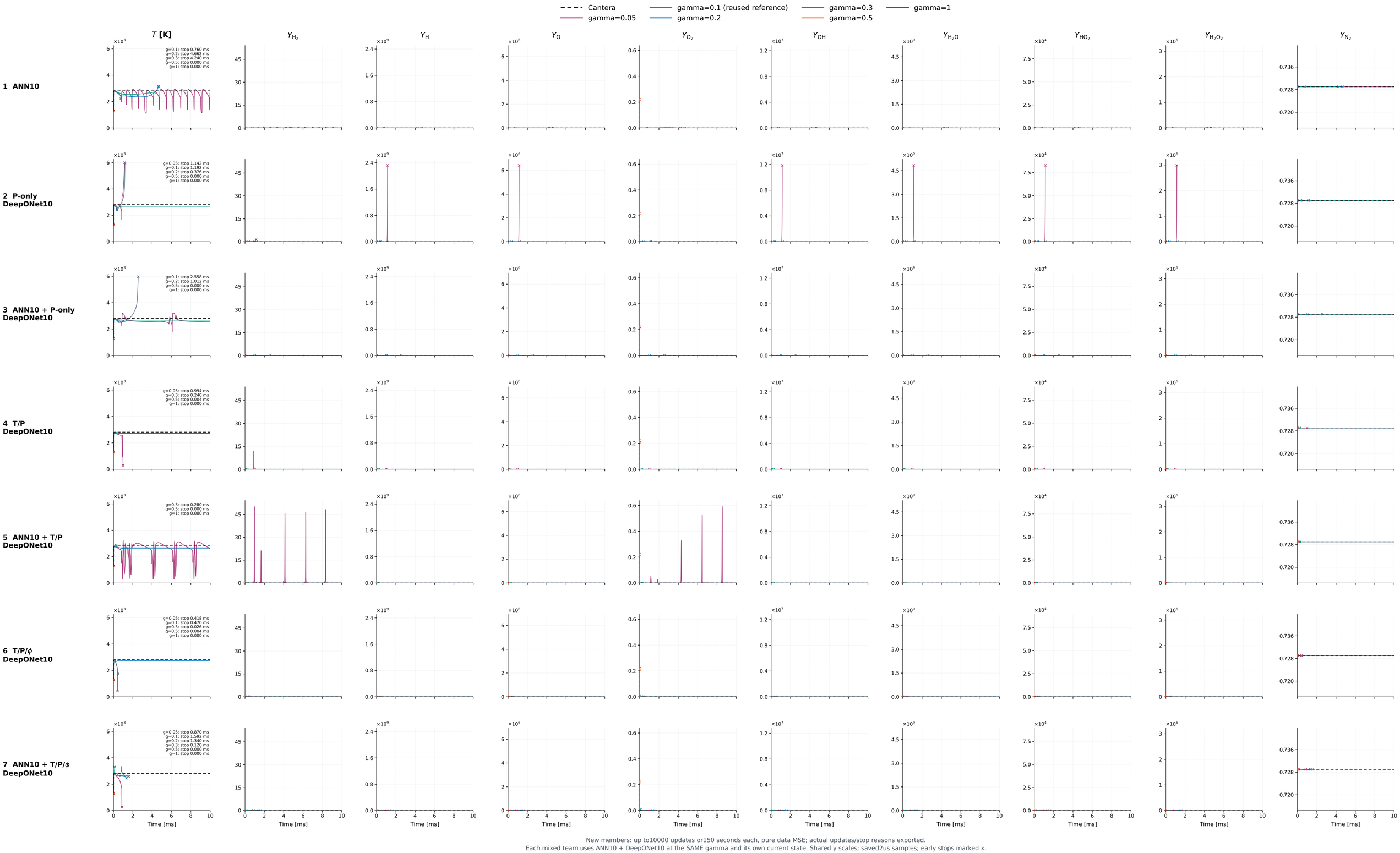
Raw | Cantera vs NN NRMSE | T0=1549.8K, P0=1.296atm, phi0=1.887

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 4.650 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.018 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 3.756 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.516 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 1.972 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.668 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.936 ms
2 P-only DeepONet10	0.2	0.133	0.429	0.394	0.694	0.144	0.663	0.102	0.281	2.61	7.64e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.406 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.092 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.558 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.892 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.678 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.444 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.768 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.486 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.042	0.0709	0.0679	0.238	0.0797	0.179	0.134	0.233	0.363	7.64e-16	Complete
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.922 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.734 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.518	0.85	5.43	9.64	77.5	4.05	2.48	1.37	1.85	1.07e-15	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.720 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.268 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.132 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.294 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.548 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 75

T_0=1274.3 K P_0=3.589 atm ϕ_0=1.775 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 75

Raw | Cantera vs NN NRMSE | T0=1274.3K, P0=3.589atm, phi0=1.775

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.156	7.88	0.736	1.95	0.0755	16.3	21.2	0.36	3.04	1.52e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.760 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 4.662 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 4.240 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.142 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.192 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.376 ms
2 P-only DeepONet10	0.3	0.0441	0.0125	0.33	0.578	0.215	0.403	0.0272	0.436	1.29	0	Complete
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.0823	1.11	2.6e+03	169	0.098	368	2.5e+03	1.57e+03	1.73e+05	6.09e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.558 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.012 ms
3 ANN10 + P-only DeepONet10	0.3	0.0462	0.0107	0.396	0.58	0.118	0.43	0.0242	0.209	0.683	9.14e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.994 ms
4 T/P DeepONet10	0.1	0.0321	0.0212	0.312	0.539	0.421	0.341	0.0198	0.784	0.55	1.37e-15	Complete
4 T/P DeepONet10	0.2	0.0349	0.0302	0.331	0.613	0.539	0.374	0.0277	0.953	0.707	0	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.240 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.226	213	305	7.58e+03	7.04	1.16e+03	44.9	2.36	12.4	6.09e-16	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0496	0.0388	0.581	0.795	0.336	0.588	0.217	0.665	0.488	1.22e-15	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0706	0.0248	0.54	0.805	0.406	0.639	0.0643	0.819	0.685	1.37e-15	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.418 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.470 ms
6 T/P/ϕ DeepONet10	0.2	0.0283	0.0288	0.289	0.545	0.543	0.314	0.0192	0.953	0.725	0	Complete
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.870 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.592 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.340 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.120 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Val IC 81

T_0=1039.6 K P_0=1.219 atm ϕ_0=1.500 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Val IC 81

Raw | Cantera vs NN NRMSE | T0=1039.6K, P0=1.219atm, phi0=1.500

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.868	2.85	0.994	0.982	3.35	1	0.999	0.661	0.859	4.53e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.260 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.722 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.526 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.130 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.874 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.100 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.336 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
4 T/P DeepONet10	0.1	0.472	0.941	3.19e+06	4.35e+05	20.6	7.53e+06	3.32e+05	9.97e+03	24	1.06e-15	Complete
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.730 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.738 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.094 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.164 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.715	5.45	2.64e+04	7.68e+04	2.34e+03	1.40e+04	672	1.33e+04	434	9.07e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.192 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.